

Human–Robot Co-Adaptation in Month-Long In-Home Mealtime Assistance

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END-TO-END MEALTIME ASSISTANCE meal 20 · Aug 13 · 1 h 02 min · 72 total skill executions



LONG-TERM IN-HOME DEPLOYMENT one month · Jul 17 – Aug 17, 2026 · 22 meals · six settings

Figure 1: *Top:* one meal end to end, run autonomously by a mobile manipulator in the care recipient’s home: retrieving the plate from the refrigerator, heating it, feeding, drinking, mouth wiping, and clearing the plate to the sink (meal 20; 72 skill executions in 62 minutes). *Bottom:* the six settings she ate in over the month-long deployment. We planned three; she proposed the other three as she grew comfortable with the robot, and the robot, in turn, learned her preferences in each.

Abstract

Physical caregiving robots hold promise for improving the quality of life of millions who require assistance with Activities of Daily Living. They must operate in people’s homes over the long term, yet most in-home studies are short-term, and longer deployments of physically assistive robots have been either teleoperated or autonomous with fixed behavior, so only the person could adapt. In this work, we deploy a mobile manipulator for a month in the home of a care recipient with Multiple Sclerosis and show human–robot co-adaptation: the robot adapts to her and she to it. The robot can provide mealtime assistance autonomously, from retrieving the meal from the refrigerator and heating it in the microwave to feeding, drinking, and mouth wiping to clearing the plate. It can do so with no input from her, or take her input at various levels, from a preference correction up to full teleoperation. She eats with it alone, with the TV on, with music, or games on her iPad, on video calls with family, and with a friend. Our key insight is that preferences are correlated, so one correction informs others; a large language model predicts her preferences jointly from her history and, after each correction, re-predicts those she has not yet confirmed, requiring 27% fewer corrections in retrospective evaluation than the strongest baseline. Her corrections fall to near zero by the second week, resurface as she becomes more comfortable with the robot and starts exploring new behaviors and her preferences change, and fall again as the robot adapts through this non-stationarity. She adapts as well: when talking at social meals confuses the robot’s open-mouth detector, she switches to a button, and by the final

day learns to talk with her mouth slightly open, opening it wide only for a bite. That day, over a video call, one correction is all the meal needs; the robot revises five other predictions from it, all matching hers. Over the month, her reported workload falls and her trust in the robot rises to the top of the scale. Website: <https://table-for-two.github.io>

1 Introduction

“I can’t remember feeding myself. Feeding yourself is normal, and you don’t remember normal. What I remember is first being fed by a caregiver at the hospital, when I was diagnosed. She mixed my pasta in with my green beans. I said I like them separate and got the ugh, what a pain. I felt like I was bossing people around, so I learned to eat it mixed.”

Babies get fed. In restaurants I used to face the wall so people couldn’t tell. When the building had a Christmas party downstairs, I didn’t go.

Now I don’t care anymore. I just eat what they give me and complain about it after. But if I’m awake at two in the morning and want a snack, I don’t wake anybody. They feed me at the rate they think I should be fed, and I’ve been done chewing, and I wait, and I wait, and I wait. Or I’m in the middle of telling a story and the fork is at my face. I’m talking.

There’s no control, because otherwise they’d have to stare at me and watch me chew. And it’s unfair to expect them to do that.

What I miss most is eating at my own pace.”

— the care recipient

Millions of people need assistance with Activities of Daily Living every day [76]. Progress in mobile manipulation [42, 48, 68], physical human-robot interaction [50, 79, 81], and robot learning with foundation models [21, 43, 45, 61, 62] is bringing autonomous robots closer to providing it in their homes. Yet most in-home deployments of autonomous physically assistive robots run for a single session or a few days [36, 37, 52]. These reveal whether a system works. They cannot reveal what happens over weeks of real use, when novelty fades [9], contexts diversify, and the relationship between a user and a robot begins to evolve.

This evolution is not one-sided. The robot must track preferences that shift with context and over time. Unlike short-term settings, long-term deployment exposes it to the full diversity of a user's life, demanding efficient personalization across many contexts with minimal workload on the user. Assistive technology that does not fit its user tends to be set aside [60]. At the same time, users accumulate experience with the robot: they learn its capabilities and failure modes, discover when to intervene, and adapt their behavior and environment accordingly. Following prior work on mutual adaptation [55, 57, 78], we call this bidirectional process *co-adaptation*, and study it as a defining feature of long-term human-robot deployment. In prior long-term deployments, however, only one side could change: physically assistive robots [53] that stayed in homes for months were teleoperated [11, 15] or ran pre-programmed motions [54], leaving adaptation to the user alone.

In this work, we study co-adaptation through **the first month-long in-home deployment of an autonomous physically assistive robot**, with a community researcher (CR) who has Multiple Sclerosis and co-designed the system under *community-based participatory research* (CBPR) [29]. Our system operates on a mobile manipulator and executes the **full mealtime-assistance pipeline autonomously**: retrieving a meal from the refrigerator, heating it in the microwave, placing it on the table, assisting the user with feeding, drinking, and mouth wiping, and dropping the plate in the sink (Fig. 1, Section 3). A pipeline execution takes about an hour, so preferences surface across many subtasks and failures accumulate. The CR can let a meal run with no input, or intervene at various levels, from confirming an action or correcting a preference to teleoperating the base and the arm. To support the robot's side of adaptation, we propose a **personalization framework** (Section 4) grounded in the hypothesis that user preferences are correlated, such that correcting one preference carries signal about the others. A large language model predicts her preferences jointly from her interaction history before each meal and, after each correction, re-predicts those she has not confirmed. Over the month, the CR eats 22 meals with the robot, one per weekday, in various scenarios: alone, with the TV on, with music, with games on a tablet, on video calls with family, and with a friend at the table.

We find that **both sides adapt over the deployment** (Section 5). On the robot's side, her corrections fall to near zero by the second week, resurface as she becomes more comfortable with the robot, starts exploring new scenarios, and her preferences change, and fall again as the robot adapts through this non-stationarity. In a retrospective evaluation on the logs, joint prediction requires 27% fewer corrections than per-dimension prediction. At the final meal, over a video call, one correction leads the robot to revise five other predictions, all matching hers. She adapts as well, changing her own

behavior to fit the robot. When talking at social meals confuses its open-mouth detector, she first switches to a button, then changes how she talks: by the final meal she speaks with her mouth slightly open and opens it wide only to request a bite. She learns not to over-exert herself, opening only as wide as the detector needs, and to engage more with her surroundings while the robot works: "I barely noticed the robot, I was looking up only when it was time to take a bite." She comes to anticipate its failures: having noticed that it is sensitive to lighting, she has the lights turned on as a room darkens, and on one day, wearing red, she asks the robot to check its plate detections with her before acting, in case it confuses her with the plate attachment.

Across the month the robot executed 1,335 skills; of the 958 that depend on perception, 93.2% ran without her input and 97.4% with her corrections or teleoperation. Her reported workload falls and her trust rises to the top of the scale; she grades the system's usability an A on the System Usability Scale [14], gives it high Technology Acceptance Model [20] scores, and rates the robot above her human caregiver on independence and control. From the month we draw **four lessons** for long-term deployment of physically assistive robots (Section 6).

2 Related Work

Physically Assistive Robots in the Home. Building on three decades of research in mealtime assistance [12, 24, 26, 35, 37, 70, 73, 77], recent deployments have fed users in their homes over multiple days: six sessions over three days [58], six meals over five days [36], and ten meals over five days [52]. These systems assist at the table; assuming a caregiver places the meal on the table, and clears it after, whereas our system assists with a meal end-to-end. Moreover, no autonomous physically assistive system has been evaluated in a person's home for more than a week. The robots that stayed longer were teleoperated or ran a fixed motion: seven days to three one-week stays of a PR2 or Stretch commanded through a web interface, head motion, or muscle signals [27, 56, 63, 80], episodic use of a PR2 across a year [15], wheelchair-mounted arms their owners have driven for years [11], and commercial feeders people have used for months [33, 54]. In these deployments the robot changed only when researchers added capabilities or reconfigured the interface [56, 63, 80], so only the person could adapt. Ours is the first month-long in-home deployment of an autonomous physically assistive robot, and to our knowledge, the first in which a physically assistive robot adapts to the person it assists from her corrections while her adaptation to it is measured. (Table 1).

Long-Term Deployments of Other Robots. A robot's first week of use is not representative of the ones that follow [25, 39, 46], and Bajones et al. [9] argue that a home robot can only be benchmarked over three weeks. In such deployments the robot changes for the site or on a schedule rather than for the individual: STRANDS learned which of its site's routes were reliable [31], Robovie unlocked behaviors as interaction time accumulated [40], and Hobbit switched modes on a predetermined day [9]. Adaptation to a specific person over a month at home has been shown only by socially assistive robots, which set instructional difficulty from a child's performance [67] or learn a per-child policy online [17]. Clabaugh et al. call this personalization co-adaptive, but frame the learning from the robot's side alone, and both studies measure the child

Table 1: Deployments of physically assistive robots. We situate our work among physically assistive robots (as defined by Nanavati et al. [53]) that meet one of three criteria: deployed for at least a week or five sessions, studied while already in a person’s everyday use, or evaluated in-home as an autonomous feeding system.

ASSISTANCE PROVIDED	DEPLOYMENT			CHANGE OVER THE DEPLOYMENT	
	SETTING	STUDY DURATION	PARTICIPANTS	ROBOT’S BEHAVIOR	PERSON’S BEHAVIOR
Fetching, scratching, wiping [27, 56, 63, 80]	Home	7–12 days; 3 × 1 week [63]	1 each	Researchers added modules [56, 63, 80]	Control strategy [80]
Grasping, pouring, pressing (person owns the robot) [11]	Home	One 4-h session ^a	7 users	No	Not measured
Feeding, fixed motion [33, 54]	Facility, home	4-day trials [33]; meal observations ^a [54]	12 in trials [33]; 2 long-term users [54]	No	Not measured
Feeding ^b [52, 58]	Home	3 days, 6 sessions [58]; 5 days, 10 meals [52]	1 care recipient each	User edits a setting [52]	Not measured
Feeding, drinking, wiping [36]	Home	5 days, 6 meals	2 care recipients	Language request	Not measured
Whole mealtime (ours)	Home	One month, 22 meals	1 person with MS	Learned from corrections, transfers across dimensions	Cueing, takeovers, lighting, plate

teleoperated or preprogrammed
autonomous

^a Prior use: JACO owned ≥6 months [11]; Bestic used 14 and 5 months [54].
^b Input per bite: judging the scoop and releasing each step [58]; locating the food [52]. Ours needs input only to switch task.

only through task skills, which improved, and engagement, which stayed flat; whether the person adapts to the robot is not asked.

Co-Adaptation and Preference Learning. Co-adaptation names adaptation in both directions, the person’s to the robot and the robot’s to the person [55, 57, 78]. The person’s side alone is well documented: households reorganized routines around a vacuuming robot that neither plans nor learns [22, 23, 75], and people’s shared-control commands grew more aligned with a robot policy that never changed [7]. Where both sides change, the one study with a real robot lasts a single session [69]; or retrain a brain-machine decoder on a user’s shifting signals [59]. Nearest to our work in predicting a user’s preferences are CBTL, which predicts each preference parameter separately with an LLM [71], and FEAST, which carries a user’s last preferences forward by setting [36]; both serve as baselines in Section 5.3. We differ in predicting all dimensions jointly, so that one correction revises the others, and are the first to evaluate preference prediction over a month of in-home use, as a user’s preferences change.

3 System

Our system is a mobile manipulator that assists with a whole meal in the CR’s home: it retrieves the plate from the refrigerator, heats it in the microwave, places it on the table, assists with feeding, drinking and mouth wiping, and drops the plate in the sink (Fig. 2). Prior mealtime-assistance systems begin once the plate is on the table [13, 28, 36–38, 52, 58], and a caregiver or researcher still brings the meal and clears it afterwards; ours is, to our knowledge, the first autonomous system to span the meal end to end (Table 1). We open-source the code and hardware designs on our website.

Hardware. Our system hardware consists of the following.

Mobile Base. The base is a custom Vention [6] stand carrying the arm, tools and compute. We add two motorized high-grip drive wheels to provide traction and convert the movable stand to a base that can autonomously navigate. An Arduino [8] relays commanded wheel speeds to two motor drivers that hold them under encoder-based PID control. Two RPLidar A1 [72] sensors provide 360° 2D lidar for localization and mapping, and a ZED Mini [74] stereo camera with an integrated IMU provides visual-inertial odometry.

Robot Arm. The arm is a Kinova Gen3 [4] 7-DoF with a Robotiq 2F-85 [5] gripper and an eye-in-hand RealSense [3] RGB-D camera.

Tool Apparatus. The stand carries a plate holder, a drink holder, and a tool stand holding the feeding and wiping utensils open-sourced by FEAST [36]; the feeding utensil is instrumented with an ATI Nano25 [1] force/torque sensor. The plate is fitted with a 3D-printed, microwave-safe polypropylene plate attachment that gives the gripper a rigid, repeatable grasp and docks the plate in its holder. A mug is fitted with a similar drink attachment.

Interaction Interfaces. Besides the web interface described below, the robot cues the CR through an LED on the gripper and a speaker on the base, and she can signal the robot through a wired button at their iPad, for example to request a bite.

Emergency Stop. A remote emergency stop, wired to the researcher’s laptop, halts the arm and base.

Compute. A Lenovo Legion Pro 7i laptop [47] with an RTX 4090 GPU runs perception, planning and preference prediction; a separate Intel NUC [2] runs the arm controllers.

Navigation. During the preparation phase we build a 2D occupancy map of the CR’s home with Cartographer [34] from the two lidars, and record waypoints for five task locations: the refrigerator, the microwave, the movable table for personal and television meals, the dining table for social meals, and the sink. During deployment, Cartographer localizes against this map. Odometry comes from the wheel encoders and the yaw rate of the ZED Mini’s IMU, fused by an extended Kalman filter [51]. Visual-inertial odometry from the ZED’s cameras, our initial approach, proved unreliable in the home due to sunlight, reflections and low-texture walls. ROS move_base [64] with the Timed Elastic Band local planner [65] commands the base’s velocity along collision-free paths it plans to the waypoints.

Manipulation. The manipulation library comprises OPENDOOR and CLOSEDOOR for the refrigerator and microwave; PRESSMICROWAVEBUTTON; PICKPLATE and PLACEPLATE at the refrigerator, microwave, tables, base holder and sink; PICKTOOL and STOWTOOL for the feeding and wiping utensils; and ACQUIREBITE and TRANSFERTOOL. The last two use FEAST’s food-manipulation skills [36, 38]

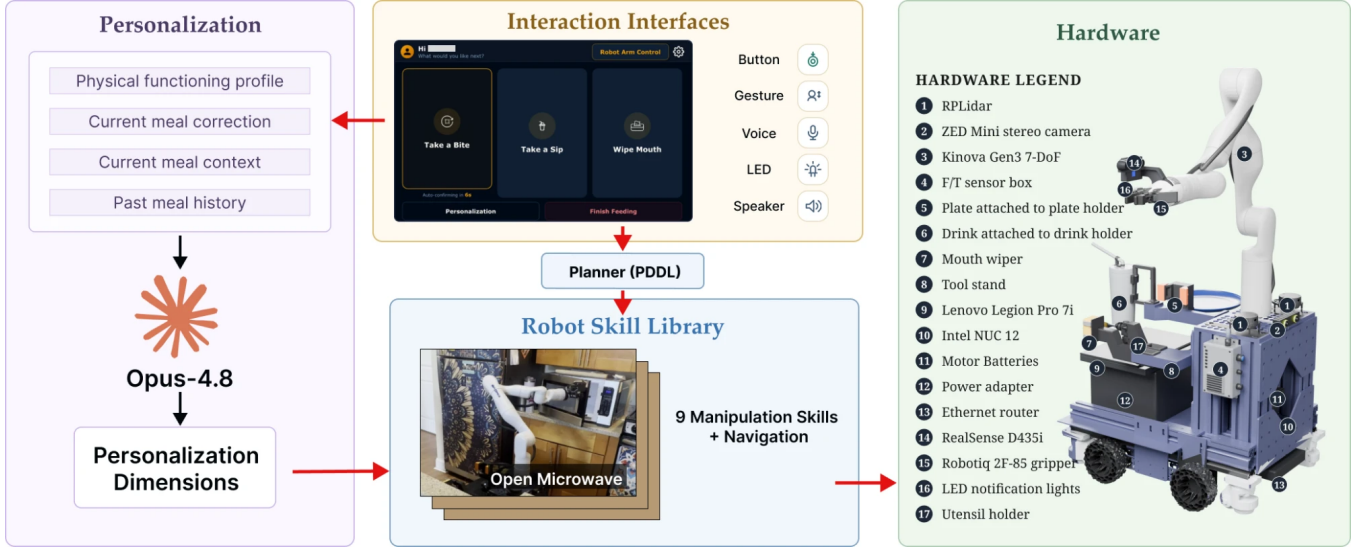


Figure 2: System overview. *Top center:* the CR requests assistance and corrects preferences through the web interface. *Left:* the personalization framework predicts her preferences jointly and re-predicts them after each correction. *Bottom center:* the predicted preferences and her inputs parameterize the navigation and manipulation skills, which run on the robot (right).

and head-pose-based transfer [37]. A PDDL task planner (FastDownward [32]) sequences these with navigation into a valid plan given the current propositional state.

The skills use 3D keypoints as the intermediate representation between perception and action, so that the CR can verify a plan before execution and correct it if necessary (see Levels of Autonomy below). The plate skills find the plate attachment by color, a per-location environment parameter the robot learns from their corrections (Section 4). The other skills use vision-language foundation models: GroundingDINO [49] detects objects such as door handles and the sink basin as open-set detections and SAM [44] refines them into masks, from which the skill takes its 2D keypoint; MolmoPoint-7B [21], prompted in natural language, returns the microwave’s start button as a point directly. The skills back-project their 2D keypoints through the RealSense depth map to 3D.

Each skill is implemented as a behavior tree [18] that anchors a waypoint trajectory on its keypoints and on typed parameters, such as $\text{Speed} \in \{\text{slow}, \text{medium}, \text{fast}\}$, that the robot predicts from the CR’s history before each meal (Section 4), and executes it through task-space and joint-space controllers.

User Interface. The web interface is a Vue.js [82] application on the CR’s iPad, communicating with the robot over rosbridge [19]. Its pages fall into four groups. *Confirmation pages* precede each step that depends on perception or ungrasps the plate: for plate detection, a color-correction page where she taps the attachment and adjusts the matched range; for the VLM keypoints, a detection page with Looks Correct or Redo; and before each plate release, a confirmation. Each auto-continues after a countdown she can set. *Preference pages* present each predicted dimension at the point it takes effect, with the predicted value preselected, and a settings overlay for revisiting any answered dimension (Section 4). *Feeding pages* build on FEAST’s open-source interface [36]: task selection, bite selection with a manual keypoint mode, and personalization

through language and gestures. *Teleoperation pages* are reachable from any page: the base page moves the robot base with a joystick, and the arm page offers joint-space and task-space control as well as predefined motions such as retract. While a skill executes, the screen shows the previous, current and next skill and a language description of what the robot is doing. Appendix A shows every page and the order in which they appear over a meal.

Levels of Autonomy. The system supports variable levels of autonomy. The CR can run a meal end to end without input: the robot predicts their preferences and, while executing the whole pipeline, auto-continues through each confirmation page after a countdown. She can also intervene at two levels. *High-level corrections* act on a skill’s inputs through the interface: correcting a predicted preference, triggering a re-detection, selecting the plate color, or manually selecting the keypoint for a bite. *Low-level teleoperation* takes over the motion itself: driving the base, for example to reposition it when navigation parks short, or commanding the arm directly to help it recover from a joint limit. She can then have the robot redo or resume the skill it was on, or move on to the next one.

Safety. Two watchdogs run continuously: one on the NUC treats the emergency stop’s signal as a heartbeat and halts the arm and base if it stops, whether from a press, a disconnected button or a lost network link; one on the laptop halts the arm if any sensor stops streaming or leaves its expected range. A collision monitor compares the arm’s joint torques with a dynamic model and stops motion when the discrepancy exceeds a threshold each skill sets for its phase, from 15 Nm to press the microwave button to 40 Nm to firmly close its door. The feeding utensil and the gripper’s 3D-printed fingertips each have a designed weak point that breaks under excess force [36]. Detections are accepted only within expected zones, and every predicted preference is checked against its parameter domain before it takes effect.

4 Personalization

Hypothesis. The robot exposes 30 *personalization dimensions*, each a choice such as how fast the arm moves or the cue she uses to request a bite. We hypothesize that they do not vary independently: how fast she wants the robot to move and how much she wants it to check with her respond to the same latent factors, such as who else is at the table, whether she is in a hurry, and how much she trusts the robot. A correction to one dimension is therefore evidence about the dimensions she has not been asked about.

Personalization dimensions. Of the 30 dimensions, 21 are *meal preferences*, drawn from requirements identified in a prior formative study with 21 care recipients [36]: five dimensions on how much the robot checks with her; eight on how she and the robot signal each other, such as whether she requests a bite by opening her mouth or pressing a button; five on the bite and how it is taken, such as how far the utensil stops from her mouth and the order in which she eats what is on the plate; two on whether the arm retracts between bites and how fast it moves; and one on how long the meal is heated. The remaining nine are *environment parameters* that describe her home, learned from her corrections in the same way: the plate attachment’s color at four pickup locations, and at five, a parking offset for the base, taken from the adjustment she makes to its position on arrival. Appendix B lists all 30 with their options.

Prediction and correction loop. The robot predicts all 30 dimensions jointly, with a single call to Claude Opus 4.8 at medium reasoning effort, from a description of the CR’s physical functioning, every prior meal’s accepted values and context in full, and this meal’s context: the food, the setting label (personal, TV or social), and the time of day. It shows the CR each predicted dimension where it takes effect (Section 3). She accepts the predicted value or replaces it, which fixes that dimension for the meal and re-predicts every dimension still open, conditioned additionally on what she has confirmed or corrected so far, so one correction can revise several others. She can also revise an already-decided dimension mid-meal from the settings overlay, which takes effect at once and likewise re-predicts the open ones. The values she ends the meal with become that meal’s record.

5 Deployment

5.1 Study Procedure

We conducted the deployment as community-based participatory research (CBPR) [29] with a community researcher (CR) who is a co-author on this paper. Our collaboration began in Fall 2022, and she has since evaluated several components of the system in her home, though never the system as a whole. She has Multiple Sclerosis, retains partial arm and neck mobility, and cannot feed herself; she operates the robot’s web interface on her iPad with her left arm. Before the study, we spend five weeks developing and testing the system in her home. Most of this time goes to navigation: the house’s narrow corridors leave little margin for error, and we make minor additions to her home, such as skirts on her tables so that the lidars see a surface rather than legs. We also plan the study with her: the meal schedule, set around her availability, and the three settings in which she expects to use the robot, eating on her own or while watching television at a movable table, and

with a friend at the dining table. Her only exposure to the complete system before the study is a single trial meal three days before it. Appendix D gives a day-by-day account of this preparation phase, including every addition to the home.

The deployment then runs for one month, July 17 to August 17, 2026: 22 meals, one per weekday, with a weekend meal making up any weekday she is unavailable. Although we plan for three settings, she proposes new ones as she grows comfortable with the robot and eats in six by the end. A researcher is present, explains the system on request, and intervenes when a skill cannot recover or safety requires it (Section 5.2). As in prior in-home deployments [10, 56, 63] we do not freeze the system: we fix it between meals as failures arise and list every change, with the meal it applied from, in Appendix F [52]. The CR chooses what she eats, when, and in what setting; every plate as served is shown in Appendix G. For every meal we collect the system’s logs, the researcher’s notes, a post-meal survey that includes what she learned about the robot that day, and a post-study survey.

5.2 Reliability in the Home

A complete meal, from taking the plate from the fridge to putting it in the sink, took a median of 59 minutes excluding researcher interventions (min 36, max 82), with medians of 15 minutes for preparation, 28 for feeding and 7 for clearing. Over the month, the robot executed 1,335 skills, 958 of them perception-dependent. For these, cumulative success was 93.2% under full autonomy, 96.8% once the CR’s high-level corrections through the interface are added, and 97.4% once her low-level teleoperation is added. Her corrections were crucial to this reliability: where perception was weakest, picking the plate up from the movable table, autonomous success was 25% and success with her high-level corrections 62.5%. We show the success rate of every perception-dependent skill in Fig. 3 and the per-meal intervention log in Appendix E. The researcher intervened 32 times across 17 of 22 meals, for a median of 1.6 minutes each, and gave 27 explanations. Time per bite, from her request through bite selection, acquisition and transfer, was 56 s at the median, against 1:00 to 2:26 for the five users and 1:41 for the community researcher in Nanavati et al. [52] over the same stages.

We changed the system between meals as failures revealed the need; Appendix F lists every change and the meal it applied from. The major one was the drinking skill, which became functional from meal 12; until then we asked her to set her drinking preferences at each meal as if the skill were available. A navigation bug that spoiled the base’s final parking at meals 8 and 13, leaving her to correct its position herself, was fixed after meal 13; at meal 13 it also cost 16 minutes of researcher intervention. Hardware repairs were confined to the gripper’s 3D-printed plastic fingertips, which worked loose or broke on three days.

5.3 Robot’s Prediction Under Non-Stationarity

The robot adapts to the CR by predicting her meal preferences and its environment parameters. Given the meal, the time of day, and the setting, personal, social, or television, it predicted all 30 personalization dimensions before each meal. Each dimension she corrected was held fixed, and the robot re-predicted every dimension still open in light of it, so one correction could change several others; we call the number of dimensions a single correction revises

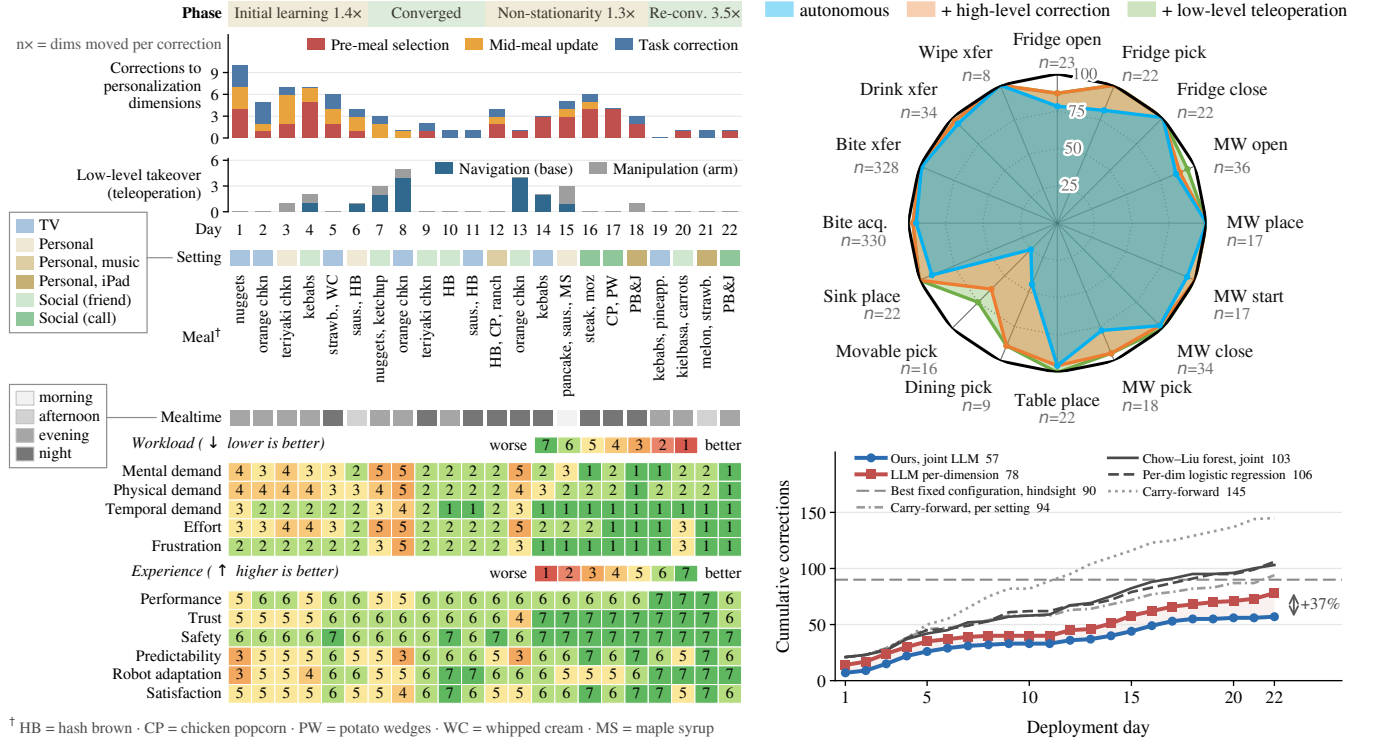


Figure 3: The deployment, day by day. Left: corrections to meal preferences and environment parameters, teleoperation, setting, meal, time and her daily ratings per meal, with the four segments of Section 5.3 marked. Top right: success rates of the perception-dependent skills. Bottom right: cumulative corrections against the baselines of Appendix C.

its *amplification*. The nine environment parameters behaved differently from the 21 meal preferences. The five parking offsets were never corrected. The four plate colors were corrected 19 times, independently: the refrigerator and microwave light their own interiors, so room lighting did not move them as a block as we had expected. The meal preferences, by contrast, were correlated, measurably: among the values she accepted, 30 of 210 dimension pairs predicted one another better than the dimension’s own most common value, against 11 by chance ($p = 0.001$, Appendix B).

Her 57 preference corrections fall into four segments (Fig. 3): Meals 1 to 6, convergence. 29 corrections as the robot learned her preferences in the three planned settings, half of them mid-meal edits as she explored what she liked and settled on it. Meals 7 to 11, plateau. 4 corrections, and none in meals 10 and 11. Meals 12 to 18, divergence. 22 corrections, for three reasons. She tried new settings as she grew comfortable: music at personal meals, a first morning meal, games on her iPad, and video calls with her daughter, which split the social setting in two (Section 5.4). Her preferences also changed: after meal 12 she wrote that “opening my mouth is not my preferred gesture for face wiping”, since unlike a bite or a sip nothing enters her mouth; at personal meals she switched the robot’s readiness cue from the LED to speech, “because I am too busy looking at the iPad screen”. And drinking became functional at meal 12, bringing its dimensions into use. Meals 19 to 22, amplified reconvergence. 2 corrections. The two descents differ in amplification: over the first six meals each of her corrections revised 1.4 other dimensions; over the final four, 3.5. The last meal is the clearest instance. On a video call with her daughter,

the robot proposed her in-person social preferences: button, LED, retract. We expected her to correct the bite trigger to open mouth and the rest of her daughter-call preferences to follow. Instead she corrected one dimension, asking to be signaled by speech, the cue she used at personal meals, and from that alone the robot revised five others, landing on her personal preferences rather than her daughter-call ones: bite and drink triggers to open mouth, the completion signal to speech, no retraction, and a countdown in place of waiting. She accepted all five. It felt natural, she said, because it was her daughter.

Over the deployment this held her correction workload to 57. Figure 3 (bottom right) replays the same 22 meals under five alternatives. The same large language model (LLM) predicting each dimension separately from its own history, as in CBTL [71], costs 78 corrections. Four count-based predictors with no LLM cost between 94 and 145: the best repeats her last accepted values in the same setting, as FEAST does [36]; the others are carry-forward, a per-dimension logistic regression, and a Chow-Liu forest over the 21 dimensions [16]. The best fixed configuration in hindsight costs 90. Only the two LLM predictors finish below it, and the joint predictor 27% below its per-dimension counterpart (Appendix C).

5.4 Human Side of Co-Adaptation

A robot that learns its user is only half of a co-adaptive system; the other half is a user who learns the robot. Over the month the CR learned what the robot perceived and when it acted, changed how she used it, and the robot adapted back.

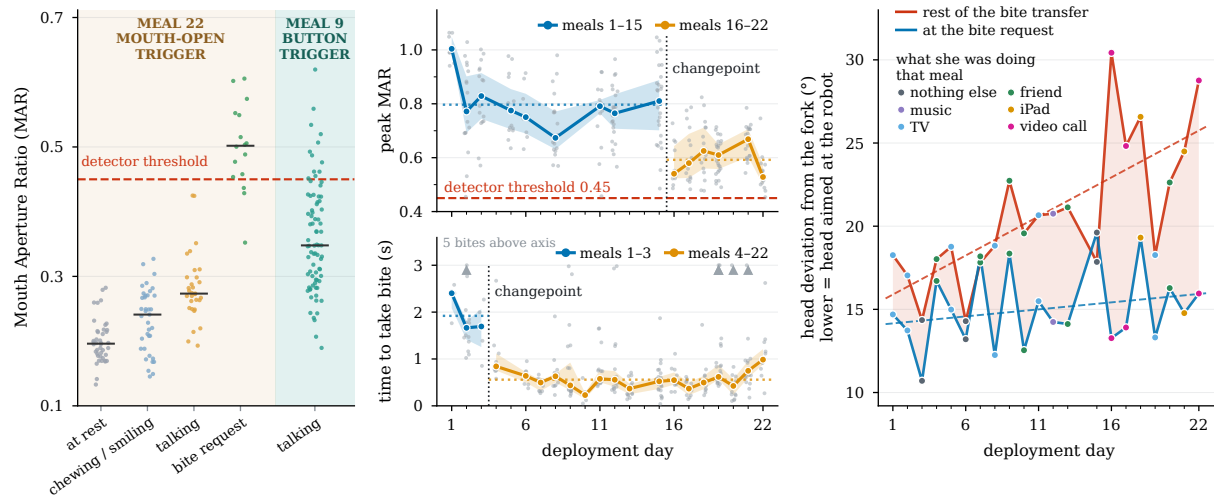


Figure 4: Four views of how the CR adapted to the robot. Left: peak MAR in each second of meal 22 and meal 9, grouped by what she was doing in that second; ticks are medians. **Center top:** her triggering aperture, which steps down at meal 16. **Center bottom:** her time to take the bite, settling by meal 4; bands are interquartile ranges, dotted lines fitted levels. **Right:** head direction relative to the fork at the bite request (± 1 s) and during the rest of the transfer (more than 3 s from it).

Learning to talk below the robot’s threshold. The open-mouth trigger fires on an aperture threshold, so it cannot tell a bite request from a word spoken with the mouth open beyond it. She met that limit at her first social meal: at meal 4, with a friend seated to her right, she switched to a button, “because I didn’t feel rushed and could carry on a conversation”, and she kept the button at her friend’s later visits, where the robot now proposed it (meals 9, 10 and 13). Meal 16 was her first video call with her daughter. Comfortable with her, she overrode the robot’s button and went back to open mouth, and began working out where talking ends and a request begins; at meal 17, her daughter on the call again, she overrode the button once more. She switched back to the button when her friend was over (meal 20, correcting the robot), but at meal 22, her third call with her daughter, she talked through the meal on the open-mouth trigger. Figure 4 (left) shows how: labeling every second of that meal’s video, not one of her 110 seconds of talking, chewing, smiling or resting reaches the 0.45 threshold, while her bite requests sit at a median MAR of 0.502 (Mann–Whitney $p = 6.4 \times 10^{-11}$). At meal 9, with the button and no reason to hold back, her talking sat at a median of 0.348 and 12 of 79 talking seconds, 15%, crossed 0.45. Her talking at the two meals is not the same distribution (Mann–Whitney on peak MAR $p = 2.3 \times 10^{-6}$; Fisher on the crossing rate $p = 0.018$): she had changed how she talked to accommodate the detector, keeping speech below the line and opening above it. “I found a way to talk and not trigger the robot to feed me a bite,” she wrote that day.

Ceasing to over-accommodate the robot. She stopped doing more than the robot needed, in two abrupt steps: from meal 16 she opened her mouth only as wide as the detector required, and from meal 4 she waited on the arriving fork only as long as she had to. Her triggering aperture over 259 bite requests sits at two levels with one step between them, at meal 16: a median MAR of 0.796 before, 0.592 from then on (Fig. 4, center top; changepoint permutation $p = 7.5 \times 10^{-4}$), 59% of her margin above 0.45 given up in a single meal, the first with her daughter on the call, where

she felt free to test the detector. Her time to take the bite once the fork arrived had settled earlier and as abruptly, from 1.92 s over the first three meals to 0.56 s from meal 4 (Fig. 4, center bottom).

Shifting engagement away from the robot. Across 357 transfers, her head direction at the moment she requested a bite, when the robot reads her head pose to aim the fork, is unchanged over the month (Fig. 4, right; $\rho = +0.24$, $p = 0.28$). During the rest of the transfer, when it does not require her head pose, she turned further from the robot as the deployment proceeded ($\rho = +0.74$, $p = 0.011$); by day 10 she would trigger a bite and carry on talking until it reached her. She chose to eat in settings that asked more of her attention, games on her iPad and video calls with her daughter on a phone propped to her left on the dining table, and changed her preferences to match, moving the robot’s readiness cue from the LED below the camera, which she was no longer looking at, to speech, which the robot learned to propose at those meals (Section 5.3). Of her last tablet meal she said, “I barely noticed the robot, I was looking up only when it was time to take a bite.”

Anticipating the robot’s failures. In the first week she learned its controls: “how to control the robot’s arm myself and restart a skill in case of an error” (day 3). In the second she learned what it perceived and where it stalled: “the robot is very sensitive to lights and shadows” (day 7); “the robot base motion may be very sensitive to my wheelchair moving around it” (day 8). In the third week she acted on the model before the robot failed. On day 14 she asked for the confirmation pages back: “I am wearing red, so I will bring back confirmation pages, as the robot might confuse me and the plate attachment”, a failure foreseen from a cue the robot cannot perceive. On day 18, after her wheelchair moved the movable table and a joint hit its limit reaching for the plate, “now I will be extra careful when moving around the movable table”, and on day 19, when she moved it again, she asked for it to be brought back within the robot’s workspace before the robot reached for the plate. On day 21, as the room darkened, she asked for the lights so the robot could still find the plate.

5.5 Rated Workload, Experience, Acceptance

After each meal the CR rated eleven 7-point items, shown day by day in Fig. 3: the five NASA-TLX workload items [30] and six on her experience of the robot. Workload fell across the month, from a mean of 2.96 in the first week to 1.36 in the last, and every item fell individually. Her experience ratings rose: trust first reached 7 at meal 14 and performance at meal 19, her rating of the robot's adaptation climbed from 3 to 7, and she never rated safety below 6. The one departure is meal 13, where trust dropped to 4 and predictability to 3 after navigation failed repeatedly, the bug fixed after that meal (Section 5.2); she said she trusted the feeding but not the driving, and both rebounded the next meal, to 7 and 6.

After the final meal she completed the System Usability Scale [14], the five Technology Acceptance Model items [20], and four items comparing the robot with her caregiver. She rated usability at 82.5, an A on the Sauro–Lewis curved grading scale [66]: she strongly disagreed that the system was cumbersome or that she would need a technical person to use it (once used to it by the end of the deployment), and strongly agreed that she would use it frequently and felt confident doing so; her one neutral rating was on whether it was unnecessarily complex. She rated usefulness, attitude, intention to use, and enjoyment 5 of 5 and ease of use 4. Against her caregiver, the robot raised her sense of independence from 2 to 5 and her sense of control from 4 to 5. She rated both sides of the co-adaptation at 5: the robot got better at understanding her preferences, and she got better at working with it.

6 Lessons Learned

Lesson 1. *The target moves: as trust grows, the user explores unforeseen scenarios and revises preferences even at familiar meals. The first plateau is not an endpoint for the learner or the study.*

The CR's corrections plateaued by meal 7 (Section 5.3), the length at which prior in-home feeding studies end [36, 52]; at that length a plateau is indistinguishable from convergence. As her trust rose (Section 5.5) she chose unplanned scenarios, ones that asked more of her attention (Section 5.4): music, video calls with her daughter, games on her tablet (meals 12 to 21). These brought new preferences: with a game on her tablet she moved the readiness cue from the LED to speech, and on the calls, comfortable with her daughter, she went back to open mouth (Section 5.4). The wiping cue moved with no change of scenario at all (Section 5.3). A learner or a study that stopped at the plateau would have missed all of it.

Lesson 2. *Preferences for an assistive robot depend on the other people present, and so does the cost of its mistakes.*

The CR chose the button with a friend at her side, and had the robot wait for her to ask for the next bite rather than count down, at every visit (Section 5.4, Appendix H); on video calls with her daughter, despite the same social label, she chose open mouth and a countdown, and at the last call one correction of hers led the robot to propose her personal preferences for the rest, which matched what she wanted; it felt natural, she said, because it was her daughter (Section 5.3). The relationship, not the label, set her preferences. At meal 20, while a friend was visiting, the button misfired and the robot moved toward her unasked, and she was more critical

of it than at any other meal. Participants in a recent speculative design study asked for a robot that adapts to the relationships at the table [41].

Lesson 3. *A user is able to tinker only when they can diagnose: variable autonomy covers the scenarios where skills fail, but the user can step in only where the robot's state is legible.*

The CR stepped in throughout the month: selecting the food skewer point herself when the detector could not separate items (25 times over 11 meals), driving the base the last stretch when navigation stopped short, holding the robot above the plate until syrup stopped dripping (day 15); Fig. 3 shows how user intervention increased success on the weakest skill from 25% to 75% (Section 5.2). She could do this where she could read the cause, lighting and shadows (day 7), her red shirt against the plate attachment (day 14), joint limits from the workspace, and by the third week she acted before failures rather than after (Section 5.4). Navigation was different: she had less information about why it failed, and it was the only capability she said she did not trust (meals 8, 13). Nanavati et al. found that users overcome off-nominals when given varied control [52]; here she had control over every component and could use it only where she could see why it was failing.

Lesson 4. *As use of a physically assistive robot grows, attention to it falls: it succeeds by fading into the background.*

During transfers the CR turned further from the robot as the month went on, while her head direction at the bite request, the moment the robot reads it, did not change (Section 5.4). Long-term studies of social robots treat a robot that fades from attention as novelty wearing off [46] and report engagement as the outcome [17]; for a robot that feeds someone, fading is the outcome. It was not disengagement: her use of the robot widened through it (Section 5.1) and her trust rose (Section 5.5). "I barely noticed the robot," she said of her last tablet meal, "I was looking up only when it was time to take a bite."

7 Discussion

In this work, we study co-adaptation through the first month-long in-home deployment of an autonomous physically assistive robot. Our results are promising: the robot learned the CR's preferences through non-stationarity, she co-adapted with the robot, and her workload fell and her trust rose over the month. However, these findings come from one person, one home, and a researcher present at every meal; longer deployments with more care recipients and less researcher presence are needed to test whether the lessons hold. Additionally, two events, a meal smoking in the microwave and a coughing fit that needed a drink at once, made it necessary for the researcher to intervene; a robot left alone would need to respond to such rare hazards faster than deliberation and more reliably than any skill here. Finally, the CR handed to the people around her not only tasks the robot could not do, such as cutting her steak, but also ones it could do but not fast enough for her, such as reheating her plate when she was in a hurry; how tasks should be allocated between an assistive robot and the care team around it is a question for future work.

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